E-Commerce Customer Segmentation & Revenue Forecasting

(Online Retail II Dataset - End-to end Data Science Project)

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1. Executive Summary

This report analyzes the Online Retail II dataset to understand customer purchasing behavior, identify customer segments, and forecast future revenue. Using RFM analysis, K-Means clustering, and SARIMA forecasting, the project delivers actionable insights that support marketing strategy, customer retention, and financial planning.

Key Findings

- Customer base naturally divides into 4 distinct behavioral segments.
- A small group of high-value customers contributes a disproportionate share of revenue.
- Revenue shows strong seasonal spikes in Early November - December.
- Forecast indicates moderate growth over the next 12 months.

This analysis provides a foundation for targeted marketing, improved customer retention, and data-driven business decisions.

2. Objectives

The objectives of this analysis are to:

- Understand customer purchasing behavior
- Build RFM (Recency, Frequency, Monetary) features
- Segment customers using K-Means clustering
- Predict customer lifetime value (CLV)
- Forecast monthly revenue using SARIMA
- Generate actionable business insights

3. Data Overview

The dataset contains the following key fields:

- Customer IDs
- Invoice Number
- Product Description (More than 4000 unique products)
- Quantity
- Price
- Invoice Dates from 2009 to 2011
- Country where transaction takes place

	Invoice	StockCode	Description	Quantity	InvoiceDate	Price	Customer ID	Country
0	489434	85048	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	2009-12-01 07:45:00	6.95	13085.0	United Kingdom
1	489434	79323P	PINK CHERRY LIGHTS	12	2009-12-01 07:45:00	6.75	13085.0	United Kingdom
2	489434	79323W	WHITE CHERRY LIGHTS	12	2009-12-01 07:45:00	6.75	13085.0	United Kingdom
3	489434	22041	RECORD FRAME 7" SINGLE SIZE	48	2009-12-01 07:45:00	2.10	13085.0	United Kingdom
4	489434	21232	STRAWBERRY CERAMIC TRINKET BOX	24	2009-12-01 07:45:00	1.25	13085.0	United Kingdom

Cleaning steps include:

- Removing canceled invoices to reflect actual purchases
- Removing negative quantities
- Removing missing customer IDs
- Product Description (More than 4000 unique products)
- Creating TotalPrice column to be used in revenue forecast = Quantity × Price
- Converting dates to datetime format to be used in analysis

4. RFM Feature Engineering

RFM metrics were computed per customer:

- Recency - number of days that passed since last purchase date
- Frequency - number of invoices per customer
- Monetary - total spending for all transactions by customer

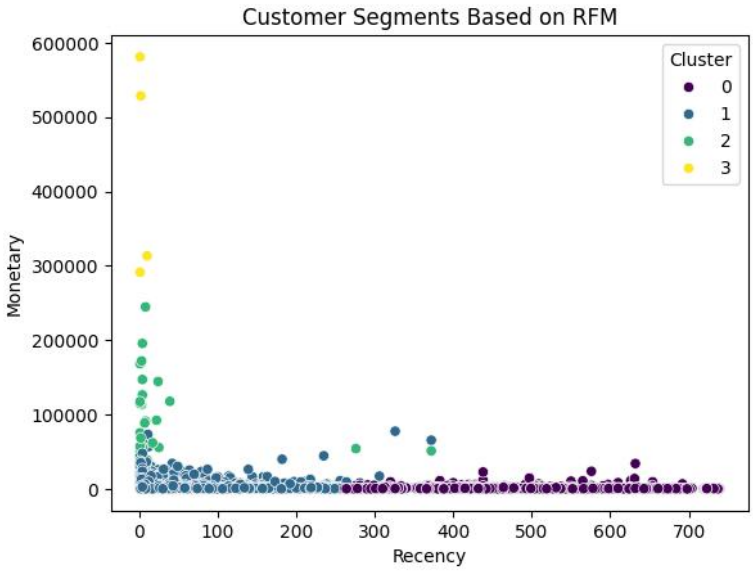
Customer ID	Recency	Frequency	Monetary
12346.0	326	12	77556.46
12347.0	2	8	4921.53
12348.0	75	5	2019.40
12349.0	19	4	4428.69
12350.0	310	1	334.40

These features summarize customer behavior and form the basis for segmentation.

5. Customer Segmentation (K-Means)

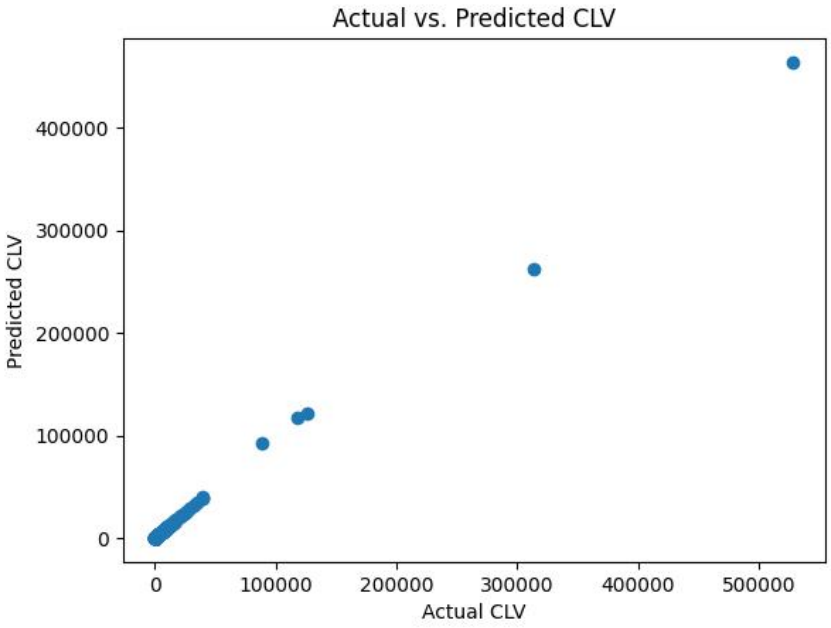
Using standardized RFM features, customers were grouped into 4 clusters:

Cluster	Description	Behavior
0	High-value loyal customers	Frequent, high spenders
1	At-risk customers	Long recency, low frequency
2	Frequent buyers	Moderate spend, high frequency
3	One-time/low-value buyers	Low frequency, low spend



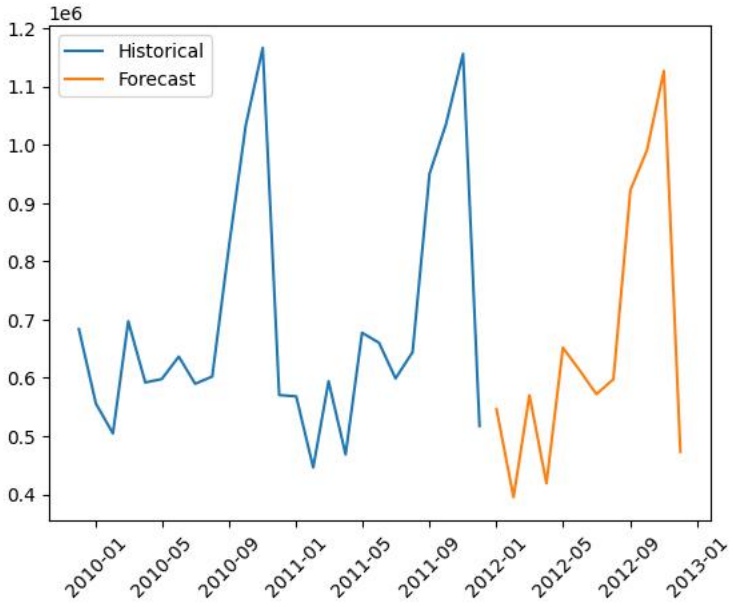
Insights

- High-value customers should be prioritized for loyalty programs.
- At-risk customers require re-engagement strategies.
- One-time buyers represent an opportunity for conversion campaigns.



6. Revenue Forecasting

Monthly revenue was aggregated and modeled using SARIMA.



Findings

- Revenue shows clear seasonality, peaking in Q4 (The fourth quarter).
- Forecast predicts moderate growth over the next 12 months.
- No major downturns are expected.

This supports inventory planning, budgeting, and marketing strategy.

7. Business Recommendations

1. **Strengthen Loyalty Programs**: High- value customers generate the most revenue. Offer exclusive discounts, early access, and personalized recommendations.
2. **Re- Engage At- Risk Customers:** Use targeted email campaigns, reminders, and limited- time offers to regain their interest.
3. **Prepare for Seasonal Peaks:** Increase inventory and marketing spend in November to December.
4. **Use Forecasting for Budget Planning:** The stable upward trend supports moderate investment in growth initiatives.

8. Conclusion

This report demonstrates how data science can extract meaningful insights from raw e- commerce data. Through segmentation and forecasting, the business gains:

- Clear understanding of customer behavior
- Actionable marketing strategies
- Predictive revenue insights